The Effect of Different Training Modalities on Force Plate CMJ Metrics

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Abstract

The purpose of this study is to look into the optimal timing for assessing countermovement jump (CMJ) metrics in order to consistently track athlete performance and development across various training modalities. Using force plates, we conducted a 13-week study with 39 collegiate baseball players, examining CMJ metrics across isometric, concentric, base strength, and power/strength combo training phases. Our goal was to determine the most effective time during a training session for performing a CMJ in order to receive the most consistent results across all training protocols. The results of this study show the timing of a CMJ during a training session does significantly influence the metrics, with the different training modalities affecting athletes performance in different ways. Simply put, isometric and base strength training showed no significant differences in pre- and post-A-block CMJ metrics, concentric training significantly improved CMJ metrics, and power/strength combo training led to a decrease in CMJ performance. This study provides information showing the importance of selecting ideal times for CMJ testing during training sessions in order to effectively track athlete performance across different training phases. Hopefully this information can help enhance training strategies and optimize athletic performance, as the insights can help coaches and sports scientists better evaluate their strength and conditioning programs.

Introduction

In sport, where split-second decisions and explosive movements win or lose games, precise evaluation of performance optimization is vital in developing effective strength and conditioning plans across all levels of athletics. Yet, tracking changes in sport specific performance across weeks and throughout seasons can be difficult due to variations in requirements for each sport. Recently, strength coaches have gravitated towards technology, such as force plates, to consistently track changes in lower body performance. While these metrics have been found to be reliable (Badby et al., 2023), consistent, optimal timing of athlete assessment has not been studied. As strength programs are designed for a different outcome and adaptation for the athlete, determining the optimal assessment window to maintain consistency over time is critical to better understand the training effects of a long term strength program.

The problem with tracking specific aspects of a training program, in a collegiate setting especially where strength coaches work with multiple sports, is there is currently no streamlined way to maintain a consistent tracking protocol across many sports. Tech companies, such as Hawkin Dynamics, have created user friendly methods which allow us to understand performance using one specific test, the countermovement jump (CMJ). The countermovement jump can be done in many varieties (ex. Hands-on-hips, single leg, arm swing, etc.) (Hawkin Dynamics Performance Team. Playbook v2b) and with the use of force plates the CMJ is measured through over 70 different metrics. The CMJ offers strength coaches with the ability to better understand their athletes power, reactive strength, jump height, fatigue, and much more in order to properly gauge the effectiveness of the program being implemented. It is through these phases that strength coaches are able to pick out key metrics they want to analyze to determine

the effectiveness of their program. Through the determination of optimal jump times, strength coaches can more effectively and efficiently analyze the progress their athletes are making and adjust accordingly.

The nuance here is these metrics need to be consistent in order to track them over a period of time. Through this study, determining the most effective jump time for consistent jump analysis throughout a lift will ensure the consistency of each metric and allow for pinpoint analysis. Especially in an off-season style training cycle where athletes are progressing towards their season, the ability to analyze consistent, strong data is crucial to ensure the program is doing as it is designed and is keeping the athletes on the correct track to be at their peak performance come time for their season. Studies have shown acute adaptations to training may affect some of the metrics outputted by the Hawkin Dynamics force plates, therefore we must question how the timing of the jump during a training session changes these outputs.

The purpose of this study was to look for consistent values in CMJ metrics in training to evaluate the most effective way to track them over a period of time. When we say “effective” or “optimal” time to jump, we are looking for the best time to jump in a training session in order to see the most consistent results across all training protocols. Throughout the 10 week study, a training program was used to build a foundation for the subjects to work off of as their season approached. Through the different phases, the study was able to uncover opportune and inopportune times to have athletes perform the CMJ. This paper offers a thorough examination of performance optimization in collegiate offseason strength training, with potential application across all aspects of peak performance within athletics. Through the study, we aim to evaluate the optimal jump time during a strength session, to consistently assess athlete progress over time. With the findings of this study, coaches can be better informed on when their athletes should jump during a training session, so they can receive the most effective and accurate results over the training period.

Methods

Subjects

The subjects of this study comprised 39 collegiate baseball players 20 +/- 1.4 years old, 87.5 +/- 8.99kg, 1.83 +/- 0.6m), highlighting the range of the study. The participants represented all baseball positions, including pitchers, infielders, outfielders, and catchers, reflecting the complete demands that are met on the field. Subjects were included in this study if they were part of the Colby College DIII Baseball team, completed 2 sessions of each type of training protocol, and completed both the pre- and post- CMJ in both sessions. Subjects were excluded if they had an injury which (A) did not allow them to lift or complete the CMJ or (B) occurred in the middle of the study which would change testing results. All study procedures were approved by the college IRB, and all participants gave informed consent. No monetary incentives were provided based upon participation.

Equipment

The equipment used for this study was the force plates by Hawkin Dynamics (Hawkin Dynamics, Westbrook, ME, USA). It consists of two portable force platforms connected to an ipad via bluetooth. These force plates measure a wide range of metrics related to performance and biomechanics, and they were used to collect the five variables of interest for this study. They are an integral part of measuring the difference in performance of a counter-movement jump before and after the A-block of a lift.

Procedures

Participants underwent a progressive strength program to prepare for an upcoming collegiate baseball season. The training blocks were progressive to prepare student athletes for the season, including breaks during week 5 and weeks 7-10 for holiday. Isometric exercises were completed weeks 1 and 2, concentric exercises weeks 3 and 4, base strength in week 6, and a power/strength combo with reinforcement exercises were completed weeks 11-13 to prepare the athletes for the season. Programs of all the different lifting phases are included in the appendix.

Hawkin Dynamics produces wireless force plates that run at 1000hz, which efficiently and effectively displays 79 metrics related to the completed countermovement jump. Participants performed countermovement jumps on the force plates before the lift began and after the initial lifting block (A-block). The jumps were specified to be completed before the A-block of the lift and immediately after the A-block was completed. There was a short warm-up before the start of the A-block, and athletes could choose whether they wanted to complete their first CMJ before or after the warm-up. This approach allowed for evaluating performance metrics at key points during the training program. The metrics used in this study included jump height, peak power output, breaking rate of force development (RFD), left-to-right (L/R) force distribution ratio, and modified reactive strength index (MRSI), providing insights into explosive strength and neuromuscular efficiency.

To complete the CMJ, a visual cue of a white light is shown on the screen, in which the subjects make a rapid downward movement (counter-movement) by flexing the knees and hips, then transitioning into an upward explosive jump as fast and high as possible. This is completed with hands on the hips at all points in time. This CMJ was completed before and after the A-block.

Metrics

Jump Height (JH): peak vertical distance a subject reaches during a countermovement jump; a measure of lower-body explosive power and force generation.

Relative Peak Power (RPP): maximum amount of power produced by the subject during countermovement jump.

Breaking Rate of Force Development (bRFD): measures how quickly an athlete can increase force production at the beginning of movement. Crucial for movements that require significant explosive strength, such as sprinting, jumping, and throwing.

The L/R force distribution ratio (LR): the balance between the left and right sides of the body during a movement. Helps identify any imbalances in strength and power between each leg.

The Modified Reactive Strength Index (mRSI): measure of subjects ability to rapidly go from the muscles being lengthened to the muscles being shortened during a countermovement jump.

Stats

Data Reduction and Analysis

Metrics were pulled from the Hawkin Dynamic Force Plates for each warm up and post-A block jump. Data lost due to a faulty test, or lack of repetition (i.e. a subject did not complete one of the two tests in a session) were excluded from further analysis.

Data are presented as mean +/- SD, p-value, and effect size (r). Shapiro-Wilk was used to check for normality. Significant changes from warmup to post-A block tests were assessed using paired t-tests, or Wilcoxon Rank Sum if non-parametric values. The p-value was set at 0.05. Statistical analyses were performed by Matlab R2023b (MathWorks, Inc; US).

Results

In the Isometric phase, the first lift saw a post decrease in mRSI from 0.58 to 0.56. This change revealed a significance at the 0.0217 level. But, in the last lift, the subjects appeared to be less fatigued during the post jumps, with bRFD from 7344.29 N/s to 7990.97 N/s ($p = 0.0178$). During this phase, all of the other metrics did not statistically change pre- to post-A-block.

Table 1: Isometric Phase Results

Phase	Lift	Metric	Pre-Value	Post-Value	Change	Significance (p-value)
Isometric	1	mRSI	0.58	0.56	-0.02	0.0217
Isometric	1	bRFD	7344.29	7990.07	+646.68	0.0178

During the base strength phase, there were no statistically different CMJ metrics pre- to post-A-block. In the concentric phase, significant improvements were noted across multiple metrics. The first lift saw a marginal increase in mRSI from 0.58 to 0.61, significant at the 0.0032 level, and a decrease in fatigue as indicated by bRFD, which improved from 7811.14 N/s to 8702.77 N/s ($p = 0.0003$). The second lift featured further enhancements: jump height increased from

0.43m to 0.45m (p = 0.0013), and bRFD, which continued to show reduced fatigue, moving from 8223.92 N/s to 9117.77 N/s (p = 0.0029). Additionally, relative power increased from 60.96 W/kg to 63.01 W/kg (p = 0.0039) and mRSI from 0.58 to 0.62 (p = 0.0134). The third lift indicated a boost in relative power from 60.97 W/kg to 62.16 W/kg (p = 0.0153). In the fourth lift, further gains were observed as jump height improved from 0.43m to 0.44m (p = 0.0429), bRFD from 7670.86 N/s to 9293.59 N/s (p = 0.0001), and relative power from 62.02 W/kg to 63.10 (p = 0.0066). mRSI also increased from 0.59 to 0.62 (p = 0.0052). These metrics collectively reveal a consistent trend of improvement post-exercise across the lifts in this phase.

Table 2: Concentric Phase Results

Phase	Lift	Metric	Pre-Value	Post-Value	Change	Significance (p-value)
Concentric	1	mRSI	0.58	0.61	+0.03	0.0032
Concentric	1	bRFD	7811.14	8702.77	+891.63	0.0003
Concentric	2	Jump Height	0.43	0.45	+0.02	0.0013
Concentric	2	bRFD	8223.92	9117.77	+893.85	0.0029
Concentric	2	Relative Power	60.96	63.01	+2.05	0.0039
Concentric	2	mRSI	0.58	0.62	+0.04	0.0134
Concentric	3	Relative Power	60.97	62.16	+1.19	0.0153
Concentric	4	Jump Height	0.43	0.44	+0.01	0.0429
Concentric	4	bRFD	7670.86	9239.59	+1622.73	0.0001
Concentric	4	Relative Power	62.02	63.10	+1.08	0.0066
Concentric	4	mRSI	0.59	0.62	+0.03	0.0052

In the Power/Strength Combo phase, the results showed a mixture of declines and improvements across different metrics and lifts. The first lift indicated a decrease in mRSI from 0.56 to 0.52 (p = 0.0481). No changes in metrics were noted in the second lift. The third lift, after instructions to increase weight, witnessed significant decreases across all measures: jump height dropped from 0.44m to 0.42m, relative power from 60.49 W/kg to 57.86 W/kg, and mRSI from 0.57 to 0.53, with significance levels at 0.0119, 0.0071, and 0.0409 respectively. The fourth lift showed continued decreases: jump height from 0.44m to 0.43m (p = 0.0009); relative power from 61.02 W/kg to 59.96 W/kg (p = 0.0088); and mRSI from 0.60 to 0.58 (p = 0.0238). In the fifth lift, similar declines were observed with jump height falling from 0.44m to 0.42m, significant at 0.0043; relative power from 60.49 W/kg to 57.86 W/kg, (p = 0.0294); and mRSI from 0.57 to 0.53, (p = 0.0472). The trend reversed in the sixth lift after the subjects were instructed to go lighter. Improvements were marked by an increase in jump height from 0.44m to 0.45m, (p =

0.0493); bRFD from 7751.81 N/s to 8569.11 N/s, (p = 0.0196); and relative power from 60.87 W/kg to 61.96 W/kg, (p = 0.013). These results encapsulate the fluctuations in performance dynamics when alternating between heavier and lighter loads.

Table 3: Power/Strength Combo Phase Results

Phase	Lift	Metric	Pre-Value	Post-Value	Change	Significance (p-value)
Power/Strength Combo	1	mRSI	0.56	0.52	-0.04	0.0481
Power/Strength Combo	3	Jump Height	0.44	0.42	-0.02	0.0119
Power/Strength Combo	3	Relative Power	60.49	57.86	-2.63	0.0071
Power/Strength Combo	3	mRSI	0.57	0.53	-0.04	0.0409
Power/Strength Combo	4	Jump Height	0.44	0.43	-0.01	0.0009
Power/Strength Combo	4	Relative Power	61.02	59.96	-1.06	0.0088
Power/Strength Combo	4	mRSI	0.60	0.58	-0.02	0.0238
Power/Strength Combo	5	Jump Height	0.44	0.42	-0.02	0.0043
Power/Strength Combo	5	Relative Power	60.49	57.86	-2.63	0.0294
Power/Strength Combo	5	mRSI	0.57	0.53	-0.04	0.0472
Power/Strength Combo	6	Jump Height	0.44	0.45	+0.01	0.0493
Power/Strength Combo	6	bRFD	7751.81	8569.11	+817.3	0.0196

Discussion

Study results show training modality acutely influences CMJ metrics, and therefore timing of the CMJ may influence decisions around athlete development and training protocols. Specifically, Base strength and isometric training showed no differences between pre- and post-A-block, post jump metrics including jump height, breaking rate of force development (bRFD), relative power, and the modified reactive strength index (mRSI) increased for concentric training and decreased after power/strength combo training.

The absence of significant differences in pre- to post-A-block CMJ metrics during the isometric training phase is one of the key findings of our study. This result somewhat aligns with the research conducted by Vargas-Molina et al. (2021), which examined the effects of isometric and isotonic exercises on vertical jump performance. Their study found isometric squats did enhance CMJ performance immediately post-exercise, but it did not produce superior outcomes compared to isotonic squats. Also, while a single set of isometric and isotonic exercises did increase post-activation performance enhancement (PAPE), a second set led to worse CMJ results. The

fact the performance enhancement was not long lasting is a possible reason for why we did not see higher CMJ performance metrics during the isometric training phase. Vargas-Molina et al. suggest the lack of dynamic movement in isometric exercise may not sufficiently mimic the actions required in a CMJ, limiting the improvements in dynamic performance metrics. This compares to concentric training which directly mimics the explosive jump of the CMJ, giving us a possible reason for why concentric training increased CMJ performance directly after training.

The basic strength training phase was another phase where we found no significant differences in pre- to post-A-block CMJ metrics. There are multiple theories behind this result. Dorrell, Smith, and Gee (2020) explored the differences in traditional percentage-based training (PBT) versus velocity-based training (VBT) on strength and power adaptations. Their findings suggest that while PBT can significantly improve maximal strength, they may not equally enhance power or explosive movements like the CMJ. Dorrell et al. also noted the significant enhancements in jump performance were observed in the VBT group. This once again matches our study as the base strength phase used PBT and concentric used VBT. Another reason for the absence of significant differences in CMJ performance can possibly be attributed to the timing of the base strength phase. This phase occurred in week 6 of the study, which was right after a week break for Thanksgiving and right before a 3 week break for finals and Winter Break. The athletes in the study may have been affected by this break.

The increase in jump metrics post-A-block for concentric training can be explained by physiological and neurological factors. Concentric training focuses on muscle shortening actions and there are several mechanisms that can explain the significant improvement in CMJ performance. Higbie et al. (1985) found concentric training improves the efficiency of motor recruitment and firing frequency. Their study suggests during concentric actions, there is a higher ratio of force to integrated electromyographic (iEMG) activity, indicating more effective activation of motor units. This enhanced neuromuscular efficiency leads to greater force production during explosive movements like the CMJ. Concentric exercises have also been shown to increase muscle activation, particularly of the agonist muscles. Folland and Williams (2007) found the rapid gains in strength observed at the start of high resistance strength training are primarily due to neurological adaptations. Adaptations including improved intermuscular coordination and motor learning can significantly enhance the muscle's ability to generate force quickly, contributing to better jump performance.

Jones et al. (2023) found compared to eccentric training, concentric training induces less muscle damage and fatigue. This allows for quicker recovery and maintains muscle performance at higher levels. Studies have shown concentric training reduces muscle twitch force to a lesser extent than eccentric training, thereby minimizing fatigue and preserving explosive power post-training session. This reduced fatigue is another possible reason for the observed improvements in CMJ metrics following the A-block of concentric training sessions.

The results of our study also show a decrease in performance metrics for the power/strength combo training phase. The findings in our study are similar to the findings by Helland et al. (2020) who examined the recovery rates from strength oriented vs. power-oriented training. They found strength-oriented training, characterized by higher loads and slower velocities, resulted in more significant neuromuscular impairment requiring longer recovery periods compared to power training sessions, which consist of lower loads at higher velocities. Greater demands for recovery could explain the decrease in performance metrics in the CMJ post-A-block, as the strength-oriented training in the phase likely introduced greater fatigue in the athletes, leading them to not be fully recovered in time for their second CMJ.

Study Limitations

This study was set up as a clinical real-world design, to evaluate the acute effects of a single bout of training on a pre-season training protocol. This off-season strength program was developed by an experienced strength and conditioning coach, with the primary purpose to prepare student-athletes for the upcoming season. Due to this real-world setting, factors such as the athlete's effort during training sessions, sleep quality and duration, and dietary habits were not directly monitored or controlled. Variations in these areas can significantly affect athletic performance including results from tests like the CMJ. Inadequate sleep or suboptimal nutrition can impair recovery leading to the athlete's performing below their peak during both training and testing sessions. Athlete motivation and effort, which are crucial in high-intensity training and performance tests, can vary and are difficult to quantify. The athletes were preparing for a high level collegiate baseball season, but fluctuations in these factors might lead to inconsistent training outcomes and CMJ performance, which can potentially skew the effects attributed to the different training phases. However, in this study a large population size and a pre-session analysis to get a baseline for each day were used in order to limit the variations in the results of this study. These measures taken were significant because the results of the study show there are still changes in the athletes CMJ's pre- and post-A-block based on the type of training the athlete is doing, which helps show this study is applicable to real life.

Practical Applications

This study has many implications for the future. It can help coaches determine the "best" time to test CMJ's on force plates during a lift by helping coaches make informed decisions about an athlete's performance based on the time jumped during a lift. In this case, "best" does not necessarily mean the highest performing metrics. We are looking for consistent values in CMJ metrics so training progress can be appropriately tracked over time. When we say "best" time to jump, we mean jump at the time during a training session where CMJ metrics will be the most consistent across sessions. Depending on the type of training being conducted it might be better to jump before or after the A-block of a lift. For example, for the best performance metrics on a CMJ it is better to jump after the A-block of a concentric training phase, but it would alter day to

day changes in performance. The results of this study can give coaches insight on why CMJ tests would significantly increase or decrease during a specific training phase. An athlete might not be getting significantly better or worse, but a specific training phase may be preparing an athlete better or making them more fatigued, skewing their CMJ's. This understanding allows coaches to optimize the timing of CMJ's so it accurately reflects the athletes performance and adaptations over the course of the training regimen.

Appendix:

Eccentric Phase Lift:

Eccentric	Reps	Tempo	Block
Force Plate Jump	2 x 1	n/a	n/a
Front Squat	5 x 3	5-0-0	A
Hurdle Hops	4 x 4	n/a	A
Trap Bar Jump	4 x 3	n/a	A
Band Assisted Vertical Jump	4 x 4	n/a	A
Band External Rotation	4 x 8	n/a	A
Force Plate Jump	2 x 1	n/a	n/a
Barbell Bench Press	4 x 4	4-0-0	B
MB Chest Pass	3 x 4	n/a	B
Chin Up	4 x 5	4-0-0	B
DB RDL	3 x 8	n/a	C
Partner Pallof Hold	3 x 4	n/a	C
Forearm Shakes	3 x 15	n/a	D
SA Overhead Tricep	3 x 10	n/a	D
MB Scoop Toss	3 x 5	n/a	D

Isometric Phase Lift:

Isometric	Reps	Tempo	Block
Force Plate Jump	2 x 1	n/a	n/a
Front Squat	5 x 3	3-2-0	A
Hurdle Hops	4 x 4	n/a	A
Trap Bar Jump	4 x 3	0-1-0	A
Band Assisted Vertical Jump	4 x 4	n/a	A
Band External Rotation	2 x 10	n/a	A
Force Plate Jump	2 x 1	n/a	n/a
Barbell Bench Press	4 x 3	3-2-0	B
MB Chest Pass	3 x 4	n/a	B
Chin Up	3 x 4	3-2-0	B
DB RDL	3 x 6	3-2-0	C
Partner Pallof Hold	3 x 10 sec	n/a	C
SA Band Tricep Pushdown	3 x 10	n/a	D
MB Hip Toss	3 x 3	n/a	D

Base Strength Phase Lift:

Base Strength	Reps	Block
Force Plate Jump	2 x 1	n/a
Front Squat	5 x 5	A
2 Hurdle + Box Jump	4 x 3	A
Band External Rotation	2 x 10	A
Band Assisted Vertical Jump	4 x 4	A
Force Plate Jump	2 x 1	n/a
Barbell Bench Press	4 x 5	B
Chin Up	4 x 7,6,5,4	B
BB RDL	3 x 6	C
Cable OH Tricep Ext	3 x 10	D
MB Overhead Throw	3 x 4	D

Power Strength Combo Phase Lift:

Power Strength Combo	Reps	Block
Force Plate Jump	2 x 1	n/a
Lateral Hurdle Hop to Broad Jump	4 x 2	A
Band External Rotation	2 x 12	A
Front Squat	5 x 3	B
Depth Jump + Jump	3 x 3	B
Force Plate Jump	2 x 1	n/a
BB Bench Press	4 x 2	C
Chin Up	4 x 8	C
Nordic Hamstring Curl	4 x 8	D
SA Band Tricep Pushdown	3 x 10	D
Tall Kneeling MB Slam	3 x 4	D

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